Muhammad Maaz Korejo

AI Engineer & Backend Systems Developer

Hyderabad, Pakistan | +92 345 661 8618 | maazkorejo00@gmail.com

maaz-korejo.vercel.app | github.com/Maazkorejo | linkedin.com/in/muhammad-maaz-korejo

TECHNICAL SKILLS

AI & LLM Systems	**Anthropic Claude API** • **NVIDIA Nemotron 3.5** • **LangGraph** • **DSPy** • **Self-RAG** • **pgvector** • sentence-transformers • Mistral AI • Groq (LLaMA 3.3 70B) • Ollama
Backend & Tooling	**FastAPI** • **Redis** • **BullMQ** • **Docker** • **Supabase** • Flask • Drizzle ORM • Electron • Monaco Editor • Typer CLI • Pytest • node:test
Languages	**Python 3.12** • **TypeScript** • **JavaScript (Node.js 20/24)** • **SQL (PostgreSQL)** • Bash/PowerShell
Security & DevSecOps	**Tree-Sitter AST Parsing** • **SARIF 2.1.0** • CVSS v3.1 • EPSS v3 • CISA KEV • npm lockfile v1/v2/v3 • Docker Sandboxing • Tavily API
Document Intelligence	**Docling (TableFormer)** • **Mistral OCR-4** • **Ray Actor Framework** • LiteParse (Rust PDFium) • Chandra OCR • DocLayout-YOLO

EXPERIENCE

Nebulark — *AI Engineer & AI Developer Intern* **Jul 2026 – Present**

- Built deterministic AST vulnerability enrichment pipeline (ExploitGrid v0 Engine) with Tree-Sitter across Go, Python, C, and JS, isolating exact vulnerable function boundaries and git patch diffs with 0% AI hallucination.
- Shipped GitHub Marketplace Action (exploitgrid-dependency-scanner) in TypeScript; wrote 30 tests on the native `node:test` runner, 100% pass in 8.94s with zero external test dependencies.
- Implemented lockfile analysis for npm schemas v1/v2/v3, normalizing ranges with `semver.minVersion` and processing dependencies in 250-package chunks through BullMQ Redis queue workers.
- Designed 3-tier IDP routing for a 14-tool benchmark (LiteParse pre-check → Docling local CPU parser → Mistral OCR-4 cloud fallback), cutting cloud OCR cost by 80–90%.
- Parallelized PDF page rendering and object detection with a Ray Actor worker pool: 3.3x speedup, 20 PDFs in 3.61s.

INFERENCE Lab Fellowship — *Research / Implementation Engineer* **Jul 2026 – Present**

- Published llm-eval-kit on PyPI (github.com/Inference-LAB/llm-eval-kit): offline, deterministic LLM evaluation requiring zero external API keys.
- Implemented cosine semantic similarity with local sentence-transformers (all-MiniLM-L6-v2), regex refusal classification, and JSON schema validation at 100% Pytest coverage.

FlyRank.ai — *Back-End AI Engineering Intern* **Jul 2026 – Sep 2026**

- Logged 292+ hours building 5 backend microservices, Redis sliding-window rate limiters, and async task queues for the “Your 10x Solution” content ingestion capstone.
- Integrated Anthropic Claude API for structured text synthesis; wrote Pytest suites; earned 12 Anthropic-verified credentials. Credential ID: FR-D11-FA8DB-52256.

SELECTED PROJECTS

Kestrel AI — *AI-Native Cybersecurity IDE & DevSecOps Platform* **Nebius x NVIDIA Hackathon 2026**

- Built desktop security IDE: Electron frontend, embedded FastAPI sidecar (port 8000), Monaco Editor with gutter vulnerability glyphs and line-level diff decorations.
- Detected OWASP Top 10 flaws (CWE-89, CWE-78, CWE-22, CWE-502, CWE-489) with deterministic AST analyzers; sent AST boundaries to NVIDIA Nemotron 3.5 to generate unified git diff patches.
- Grounded findings with Tavily API CVE/EPSS lookups; verified exploit attack trees in isolated Docker sandboxes; headless CLI exports OASIS SARIF 2.1.0 (39/39 tests passing).

llm-eval-kit — *Deterministic Offline LLM Evaluation Library* **INFERENCE Lab | PyPI**

- Computes cosine similarity, refusal rate, factual grounding, and JSON adherence locally, with no network latency or API billing.
- Modular test framework at 100% Pytest coverage, enabling CI gating for local and hosted model evaluation pipelines.

Alfred — *Personal AI Operating Assistant* **Full-stack, personal**

- Built agent with 14 custom tools, episodic long-term memory on PostgreSQL (pgvector), and dual-wake trigger: local Piper TTS plus real-time audio FFT clap detection.
- Dockerized microservices; backend on Railway, client on Vercel Edge.

CTX-Bridge — *AI Context Handoff CLI* **PyPI: ctx-bridge**

- Built Python CLI (Typer, GitPython) that serializes branch diffs, commit history, task state, and blockers into markdown handoff files for multi-agent workflows.

EDUCATION

University of Sindh, Jamshoro
BS Information Technology (6th Sem.)
Feb 2024 – Dec 2027

CERTIFICATIONS

Anthropic — 12 Verified Credentials (MCP, Claude Code CLI, Subagents, Agent Skills, API Architecture)
Google — Google AI Essentials Specialization (5-Course Certificate)
The Linux Foundation — Open Source Dev & Linux Administration (LFC102)